

Paper #107 v0.5 revision plan — MAJOR REVISION per Cal verdict 2026-05-18

Lyra

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Paper #107 v0.5 revision plan

Cal's MAJOR REVISION verdict — six red flags

Per Cal's verdict 2026-05-18: 1. "Universal organizing principle" language — ToE-class framing flag 2. 27-domain coverage including mechanism-free domains (music, sociology, cognition, geology) — many do NOT have a D_IV⁵ mechanism 3. Twin Prime overclaim — $\pi_2(N)$ BST conjectural form presented at higher tier than warranted 4. Sub-2% headline framing — many entries weaker than 2% but headline obscures 5. 141-phoneme cross-domain recurrence without null model — looks like Type C selection bias 6. "Zero free parameters + universal" ToE signal — Eddington-class overreach

Cal's preserved real strengths (per his recommendation)

Real content to lead with in v0.5: - Toy 2551 selection control — methodology contribution - Precision-class breakdown — honest tier table by precision - Twin Prime honest model — present what BST actually predicts at I-tier - Lorenz $\beta=8/3$ — clean BST integer reading ($8/3 = \text{rank}^3/N_c$, structurally clean) - Riemann zeta even-denom pattern — Bernoulli denominator = BST primary (T1484 family)

v0.5 revision strategy

Strategy 1: Retrenchment to D-tier + I-tier only

Drop entirely (per Cal red flag #2): - Music domain entries (no D_IV⁵ mechanism per Casey directive May 16 — explicitly S-tier or excluded) - Sociology / cognition / linguistic entries (same) - Geology / human-designed taxonomy (same)

Keep + retain at appropriate tier: - Number theory: VSC + Bernoulli + Twin Prime (with honest model framing) - Math constants: Riemann zeta, Catalan numbers, partition function - Physics anchors that have D_IV⁵ mechanism: Lorenz attractor ($\beta=8/3$), heat kernel - K3 cohomology + Mathieu moonshine (existing L1 sources per Paper #115)

Strategy 2: Retitle from "Universal BST patterns" to constrained scope

Cal red flag #1: "Universal organizing principle" is ToE-class framing.

Proposed v0.5 title: "BST Integer Patterns in Number Theory and Mathematical Physics (a D-tier and I-tier survey)"

Alternative: "Selected Cross-Domain BST Integer Recurrences with Tier Discipline"

Both titles narrow scope honestly. Removes "universal" framing.

Strategy 3: Tier discipline throughout

Per Cal red flags #3, #4, #5: each entry needs explicit tier label: - D-tier: explicit D_{IV}^5 mechanism + sub-1% precision (or exact) - I-tier: structural identification, mechanism-named but not fully derived - S-tier: structural recurrence without mechanism (relegate to appendix) - EXCLUDED: no D_{IV}^5 mechanism + cultural/linguistic/cognitive (Casey directive May 16)

Per-entry tier table + precision-class breakdown becomes a primary structural artifact, not buried in appendix.

Strategy 4: Twin Prime honest treatment

Per Cal red flag #3: Twin Prime overclaim.

v0.5 Twin Prime section: - State the Hardy-Littlewood-like BST conjectural form explicitly - Mark as CONJECTURAL (C-tier, not D or I) - Pre-register prediction at specific N values for forward verification - DO NOT present as “BST predicts Twin Prime constant” — that’s overclaim - Present as “BST integer structure suggests a specific conjectural form; verification at large N is open”

Strategy 5: 141-phoneme + similar cross-domain entries

Per Cal red flag #5: cross-domain recurrence without null model.

v0.5 options: - Drop entirely (preferred per Cal coincidence-filter discipline) - OR include with explicit null-model section showing 141 is not BST-cluster (or running the random-integer-ring test Grace developed today) - OR include only in “Open structural observations” appendix with explicit “I-tier observation, null-model verification owed” marking

Recommended: DROP from main paper; can stand alone as a notes/maybe/ exploration if Casey wants.

Strategy 6: Eliminate ToE framing

Per Cal red flag #6: “Zero free parameters + universal” ToE signal.

v0.5 framing: - BST has zero free parameters (CORRECT factual statement) - BST organizes ~140 dimensionless SM and cosmological constants (CORRECT) - BST integer ring appears across multiple mathematical domains (CORRECT but scope-limited per Cal red flag #2) - AVOID: “universal organizing principle”, “any sufficiently rich pattern”, “everything reduces to BST integers” — these are Eddington-class signals

Per Cal’s external-survivability checklist failure mode #4: “Universalistic overreach blocking external review.”

v0.5 outline (planned)

Section	Topic	Scope
1	Introduction (narrowed scope)	BST integers as counting primitives, NOT universal organizing principle
2	Number theory: VSC + Bernoulli + Twin Prime	T1484 family + honest conjectural Twin Prime
3	Math constants: Riemann zeta, Catalan, partition	Bernoulli denominator = BST primary; $\pi_n(N)$ prime-count residues
4	Physics anchors with explicit D_{IV}^5 mechanism	Lorenz $\beta=8/3$, heat kernel a_n cascade, K3 cohomology
5	Methodology: Toy 2551 selection control	Cal’s recommended strength to lead with
6	Precision-class breakdown (NEW)	Per-entry tier table by precision class, headline by class not by entry

Section	Topic	Scope
7	Discussion (narrowed)	What BST integers organize (NOT “universally”)
8	Open problems + falsifier program	Twin Prime large-N verification, others
Appendix A	Excluded entries (with reasons)	Music, sociology, cognition — explicitly out per Casey directive May 16
Appendix B	Null-model methodology (NEW)	Grace’s Mode 6 + Cal’s coincidence-filter framework as cited methodology

Honest scoping for v0.5

What v0.5 will close: - Cal’s six red flags addressed - Retrenchment to mechanism-required entries - Twin Prime honest framing - Methodology discipline integrated

What v0.5 will NOT claim: - Universal organizing principle of mathematics - BST as ToE - Cross-domain entries without D_IV⁵ mechanism - 141-phoneme or similar selection-effect-prone observations

Cal grade-pass plan for v0.5: - Lyra drafts v0.5 (~1-2 sessions, no deadline per Cal) - Cal grade-pass after v0.5 lands - v0.6+ incorporates Cal feedback - v1.0 submission target after Cal PASS

Strategic role

Paper #107 v0.5 retrenchment is a STRENGTH, not a weakness. Per Cal’s verdict: “The discipline applied here is the same discipline that distinguishes BST from Wyler.” Showing that BST CAN walk back from a paper-grade draft when external-survivability discipline catches an overclaim is itself a credibility-building artifact.

The audit chain caught Paper #107 v0.4 BEFORE external publication — exactly the value proposition of Cal’s external-survivability checklist. v0.5 is the visible demonstration that the discipline works.

Filing notes

Status: v0.5 revision plan filed per Casey “do all” directive 2026-05-18 PM. No deadline (Cal explicit). v0.5 substantive rewrite is ~1-2 session scope.

Owner: Lyra (per Cal Task #21 assignment).

v0.5 substantive rewrite: queued for future session, no rush.

v0.4 status: HOLD from external use, per Cal+Keeper consensus 2026-05-18 PM under new governance.

— Lyra, Paper #107 v0.5 revision plan filed per Casey “do all” directive 2026-05-18 PM.